What pitch is a growling stomach? A unified multi-mode acoustic model of borborygmi

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Abstract

Borborygmi—the rumbling and gurgling sounds of the gastrointestinal tract—arise from oscillation of gas pockets entrained in intestinal fluid. Despite their clinical significance as indicators of gut motility, no first-principles acoustic model predicts their frequency spectrum from measurable anatomical parameters. We present a unified analytical framework comprising five resonance mechanisms: (i) free Minnaert bubble oscillation, (ii) tissue-constrained bubble resonance incorporating intestinal wall elasticity, (iii) Helmholtz resonance through peristaltic constrictions, (iv) axial piston modes of cylindrical gas slugs, and (v) radial breathing modes of gas columns in elastic tubes. The constrained bubble model predicts frequencies of 135 Hz to 440 Hz for gas pocket volumes of 1 mL to 50 mL, overlapping the clinically observed 200 Hz to 550 Hz band for healthy adults without curve-fitting. A mode-transition map identifies which resonance mechanism dominates in each region of the volume–geometry parameter space. The framework provides a physics-based tool for interpreting auscultatory findings and designing acoustic monitoring systems for gastrointestinal motility assessment.

Keywords: borborygmi, bowel sounds, Minnaert resonance, bubble acoustics, gastrointestinal acoustics, Helmholtz resonator

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1. Introduction

Few acoustic phenomena are as universally experienced yet as poorly understood as the sounds of the gastrointestinal tract. Borborygmi—the medical term for stomach and intestinal gurgling—have been documented since at least Cannon's [1] pioneering auscultation studies, and their clinical significance as indicators of gut motility, obstruction, and post-operative recovery is well established [2, 3, 4]. Despite this long history, clinical assessment of bowel sounds remains largely qualitative: present or absent, normoactive or hyperactive, high-pitched or low-pitched.

The past two decades have seen growing interest in quantitative bowel sound analysis. Craine et al. [3, 5] developed computerised auscultation systems and reported spectral differences between normal and obstructed bowel. Ching and Tan [2] performed spectral analysis using electronic stethoscopes and documented that healthy bowel sounds occupy a band of approximately 200 Hz to 550 Hz with a peak near 340 Hz, while intestinal obstruction shifts the spectrum in condition-specific ways. More recently, Nowak et al. [6] reviewed automated bowel sound analysis using machine learning classifiers, and Dalle Donne et al. [4] surveyed wearable monitoring systems. These approaches treat bowel sounds as signals to be classified, but offer no predictive model explaining *why* particular frequencies appear.

The physics of bubble acoustics, by contrast, is mature. The classical result of Minnaert [7] for the resonant frequency of a spherical gas bubble in an incompressible fluid remains the starting point for underwater and biomedical acoustics alike [8]. Church [9] extended the theory to bubbles encapsulated in elastic shells—primarily for ultrasound contrast agents—and Hoff [10] provided a comprehensive treatment of coated bubble dynamics. Li et al. [11] recently derived rigorous Minnaert corrections for bubbles in soft elastic media, while Ammari et al. [12] analysed bubbly media from a mathematical perspective. None of these works, however, has been applied to the specific geometry, material properties, and clinical context of the gastrointestinal tract.

We bridge this gap. In this paper we present a unified analytical framework in which intestinal gas pockets are modelled as acoustically resonant systems, with five distinct mechanisms producing frequencies that overlap the clinically observed spectrum. The model requires no fitted parameters: all inputs are measurable anatomical and material properties—gas pocket volume, intestinal wall stiffness, lumen diameter, and constriction geometry—drawn from published literature or standard anatomical measurements. The tissue-constrained bubble mechanism was first analysed in the context of abdominal resonance by [13], whose constrained-bubble formulation we extend here to the intestinal geometry. The tissue-constrained bubble model, which augments the Minnaert formula with elastic wall stiffness and inertia, captures much of the healthy bowel sound band of 200 Hz to 550 Hz for pocket volumes of 1 mL to 50 mL, specifically the 200 Hz to 440 Hz sub-band. Frequencies above 440 Hz likely require sub-millilitre pockets or Helmholtz contributions. A mode-transition map reveals which resonance mechanism dominates in each region of the volume–diameter parameter space, providing physical insight for interpreting auscultatory findings.

2. Theory

We model a gas pocket of volume V trapped within the fluid-filled lumen of the intestinal tract. The pocket is surrounded by incompressible intestinal fluid of density $\rho_f = 1020\,\mathrm{kg/m^3}$ and confined by an elastic intestinal wall of Young's modulus E, Poisson's ratio ν, thickness h, and density ρ_w. Five resonance mechanisms are considered, each corresponding to a distinct physical geometry and oscillation pattern. Table 1 summarises the canonical parameter values used throughout.

Table 1: Canonical parameter values for the borborygmi model.

Parameter	Symbol	Value	Unit
Gas adiabatic exponent	γ	1.4	—
Atmospheric pressure	P_{atm}	101,325	Pa
Intra-abdominal pressure	P_{iap}	1,000	Pa
Ambient pressure	P_0	102,325	Pa
Intestinal fluid density	ρ_f	1,020	kg/m^3
Intestinal wall modulus	E	10	kPa
Wall Poisson's ratio	ν	0.45	—
Wall thickness	h	3.0	mm
Wall density	ρ_w	1,040	kg/m^3
Lumen diameter	d	30	mm
Gas sound speed	c_g	343[a]	m/s
Neck diameter (Helmholtz)	d_n	5	mm
Neck length (Helmholtz)	L_n	10	mm
Loss tangent	η	0.25	—

[a]Room-temperature value; air at 37 °C gives $c_g \approx 353\,\mathrm{m/s}$, a ∼3% difference that affects only the Helmholtz mode and is smaller than the parametric uncertainty.

2.1. Free Minnaert bubble

The classical Minnaert frequency [7] for radial pulsation of a spherical gas bubble of radius R in an unbounded incompressible fluid is

$$f_M = \frac{1}{2\pi R}\sqrt{\frac{3\gamma P_0}{\rho_f}}, \tag{1}$$

where γ is the polytropic exponent, P_0 is the ambient pressure, and ρ_f is the fluid density. The equivalent spherical radius is related to the pocket volume V by $R = (3V/4\pi)^{1/3}$. For a 10 mL pocket ($R = 13.4\,\mathrm{mm}$) in intestinal fluid at body temperature, Eq. (1) gives $f_M = 244\,\mathrm{Hz}$.

The Minnaert frequency provides a useful baseline but neglects the elastic

constraint of the surrounding intestinal wall. For a bubble suspended in free fluid this is appropriate; for a gas pocket enclosed within the gut, it is not.

2.2. Tissue-constrained bubble

The intestinal wall wrapping the gas pocket contributes both elastic stiffness and inertia. Following the approach of Church [9] for encapsulated bubbles, we model the wall as a thin elastic shell of thickness h concentric with the spherical gas pocket. The restoring force per unit surface area per unit radial displacement comprises a gas compressibility term and a wall elasticity term:

$$k_{\text{gas}} = \frac{3\gamma P_0}{R}, \qquad k_{\text{wall}} = \frac{2Eh}{R^2(1-\nu)}, \tag{2}$$

where E and ν are the wall Young's modulus and Poisson's ratio, respectively. The effective inertia per unit area consists of the radiation added mass of the surrounding fluid and the shell mass:

$$m_{\text{fluid}} = \rho_f R, \qquad m_{\text{wall}} = \rho_w h. \tag{3}$$

The undamped natural frequency is then

$$f_c = \frac{1}{2\pi}\sqrt{\frac{k_{\text{gas}} + k_{\text{wall}}}{m_{\text{fluid}} + m_{\text{wall}}}}. \tag{4}$$

In the limit $E \to 0$ and $h \to 0$, the wall contributions vanish and Eq. (4) reduces exactly to the Minnaert result, Eq. (1). We have verified this reduction both analytically and numerically.

For the soft intestinal wall ($E = 10\,\text{kPa}$, $h = 3\,\text{mm}$, $\rho_w = 1040\,\text{kg/m}^3$), wall inertia m_{wall} dominates over wall stiffness k_{wall}, and the constrained frequency is *lower* than the free Minnaert value. At 10 mL, $f_c = 223\,\text{Hz}$ compared with $f_M = 244\,\text{Hz}$—the wall adds mass without proportionate stiffness. This physically intuitive result (the gut wall is heavy and compliant) means the constrained model shifts predictions *towards* the lower end of the clinical band, improving agreement.

Damping is introduced through the loss tangent η of the wall material, giving a quality factor $Q = 1/(2\zeta) = 1/\eta = 4.0$ and a damping ratio $\zeta = \eta/2 = 0.125$.

2.3. Helmholtz resonator

Peristaltic contractions can create a constriction—or "neck"—connecting the gas pocket to the adjacent lumen, forming a Helmholtz resonator [14, 15]. The resonant frequency is

$$f_H = \frac{c_g}{2\pi}\sqrt{\frac{A_n}{L_{\text{eff}} V}}, \tag{5}$$

where c_g is the speed of sound in the gas, $A_n = \pi d_n^2/4$ is the neck cross-sectional area, V is the pocket volume, and $L_{\text{eff}} = L_n + 1.6\, r_n$ is the effective neck length including the Rayleigh end correction [14], with $r_n = d_n/2$ the neck radius.

For a 10 mL pocket with a 5 mm neck diameter and 10 mm neck length, $f_H = 647\,\text{Hz}$. This lies above the typical healthy bowel sound range but within the spectrum reported for obstructed bowel [2, 16], suggesting that Helmholtz resonance may contribute to the high-pitched "tinkling" sounds characteristic of intestinal obstruction.

2.4. Axial piston mode

When the gas pocket assumes a cylindrical slug geometry filling the lumen cross-section, it can oscillate axially between fluid plugs on either side. The gas slug of length $L_s = V/(\pi R_l^2)$, where $R_l = d/2$ is the lumen radius, acts as a compressible spring, while the fluid plugs provide inertia through their radiation added mass:

$$k_{\text{ax}} = \frac{\gamma P_0 \pi R_l^2}{L_s}, \qquad m_{\text{end}} = \frac{8}{3}\rho_f R_l^3, \tag{6}$$

where m_{end} is the radiation mass of each fluid piston face [15]. With two identical end masses, the natural frequency is

$$f_{\text{ax}} = \frac{1}{2\pi}\sqrt{\frac{k_{\text{ax}}}{2\, m_{\text{end}}}}. \tag{7}$$

For a 10 mL slug in a 30 mm tube, $f_{\text{ax}} = 99\,\text{Hz}$—well below the audible borborygmi range. This mode likely contributes to the sub-audible low-frequency rumbling component of gut motility.

2.5. Radial breathing mode

A cylindrical gas column confined within an elastic tube can undergo radial breathing oscillations, in which the gas and tube wall expand and contract together. The effective stiffness per unit length includes gas compressibility and wall elasticity:

$$K_{\text{gas}} = 2\gamma P_0, \qquad K_{\text{wall}} = \frac{Eh}{R_l(1-\nu^2)}, \tag{8}$$

where the factor 2 (rather than 3 for the sphere) reflects cylindrical geometry. The effective mass per unit length is

$$M_{\text{fluid}} = \rho_f R_l, \qquad M_{\text{wall}} = \rho_w h, \tag{9}$$

and the natural frequency is

$$f_r = \frac{1}{2\pi R_l}\sqrt{\frac{K_{\text{gas}} + K_{\text{wall}}}{M_{\text{fluid}} + M_{\text{wall}}}}. \tag{10}$$

For a 30 mm tube, $f_r = 1329\,\text{Hz}$. This frequency is largely independent of gas pocket volume (depending only on tube properties) and lies above the typically reported bowel sound range. The radial mode may contribute to high-frequency components occasionally observed in colonic recordings, where larger lumen diameters would bring f_r into the audible range.

2.6. Mode selection and coexistence

Real intestinal gas pockets are irregular in shape and may simultaneously exhibit characteristics of several idealised geometries. The pocket geometry determines which mode dominates: quasi-spherical pockets favour the constrained bubble mode; elongated slugs excite axial piston oscillations; and pockets connected to the lumen through peristaltic constrictions act as Helmholtz resonators.

We define the dominant mode at each volume as the one producing the lowest frequency, on the physical grounds that the lowest-frequency mode requires the least energy to excite and is most readily driven by broadband peristaltic forcing. For the canonical parameters, the constrained bubble mode has the

lowest frequency below approximately 0.22 mL; above this volume, the axial piston mode takes over as the lowest-frequency mode because its frequency drops rapidly with increasing slug length ($f_{\mathrm{ax}} \propto V^{-1}$ vs. $f_c \propto V^{-1/3}$ for the constrained bubble).

However, the axial piston frequency falls below the audible range at physiological volumes: $f_{\mathrm{ax}} < 100\,\mathrm{Hz}$ for $V > 1\,\mathrm{mL}$. Consequently, the constrained bubble mode dominates the *clinically audible* band (200 Hz to 550 Hz) across the entire physiologically relevant volume range, even though it is not the overall lowest-frequency mode.

The coexistence of multiple modes at each pocket volume provides a natural explanation for the broadband character of borborygmi: a single peristaltic event compressing a gas pocket excites several resonance mechanisms simultaneously, producing a sound with contributions at multiple frequencies.

3. Results

3.1. Canonical predictions

Table 2 presents the predicted resonant frequencies for all five modes with a 10 mL gas pocket in a 30 mm diameter intestinal segment. The constrained bubble frequency of 223 Hz falls within the healthy bowel sound band (200 Hz to 550 Hz), while the Minnaert baseline at 244 Hz is also plausible. The Helmholtz mode at 647 Hz corresponds to higher-pitched obstruction sounds, and the axial piston mode at 99 Hz to sub-audible rumbling. The radial breathing mode at 1329 Hz is rarely reported clinically.

3.2. Volume dependence

Figure 1 shows all five mode frequencies as a function of gas pocket volume from 1 mL to 50 mL, with the clinical bowel sound band (200 Hz to 550 Hz) indicated by shading. All bubble- and slug-based modes decrease monotonically with increasing volume, following the expected $f \propto V^{-1/3}$ scaling for the Minnaert-type modes and $f \propto V^{-1/2}$ for the Helmholtz and axial modes. The

Table 2: Predicted resonant frequencies for a 10 mL gas pocket in a 30 mm diameter intestinal segment.

Mode	Expression	Frequency (Hz)	Clinical relevance
Free Minnaert	Eq. (1)	244	Baseline estimate
Constrained bubble	Eq. (4)	223	Within healthy range
Helmholtz resonator	Eq. (5)	647	Obstruction / tinkling
Axial piston	Eq. (7)	99	Sub-audible rumbling
Radial breathing	Eq. (10)	1329	Rarely reported

radial breathing mode is independent of gas pocket volume as it depends solely on tube properties.

The constrained bubble model covers 135 Hz to 440 Hz over this volume range, overlapping the lower portion of the clinically observed healthy bowel sound band. At $V = 1\,\mathrm{mL}$, the constrained frequency is 440 Hz; at $V = 50\,\mathrm{mL}$, it decreases to 135 Hz. The Minnaert model predicts a slightly wider range of 143 Hz to 527 Hz, but systematically overpredicts relative to the constrained model because it neglects wall inertia.

3.3. Mode transition map

Figure 2 presents a mode transition map showing all five mode frequencies on logarithmic axes across a volume range of 0.1 mL to 100 mL. Crossover points mark volumes at which the lowest-frequency mode changes. For the canonical parameters, a single crossover occurs at $V \approx 0.22\,\mathrm{mL}$ ($f \approx 667\,\mathrm{Hz}$), where the constrained bubble mode (dominant at very small volumes) yields to the axial piston mode, whose frequency decreases more steeply with volume.

Above this crossover, the axial piston mode is formally the lowest-frequency mode across the physiologically relevant range (1 mL to 50 mL). However, f_{ax} is sub-audible at these volumes ($< 100\,\mathrm{Hz}$ for $V > 1\,\mathrm{mL}$), so the constrained bubble mode is the dominant mechanism within the *clinically audible* band, confirming it as the primary contributor to auscultated bowel sounds. The

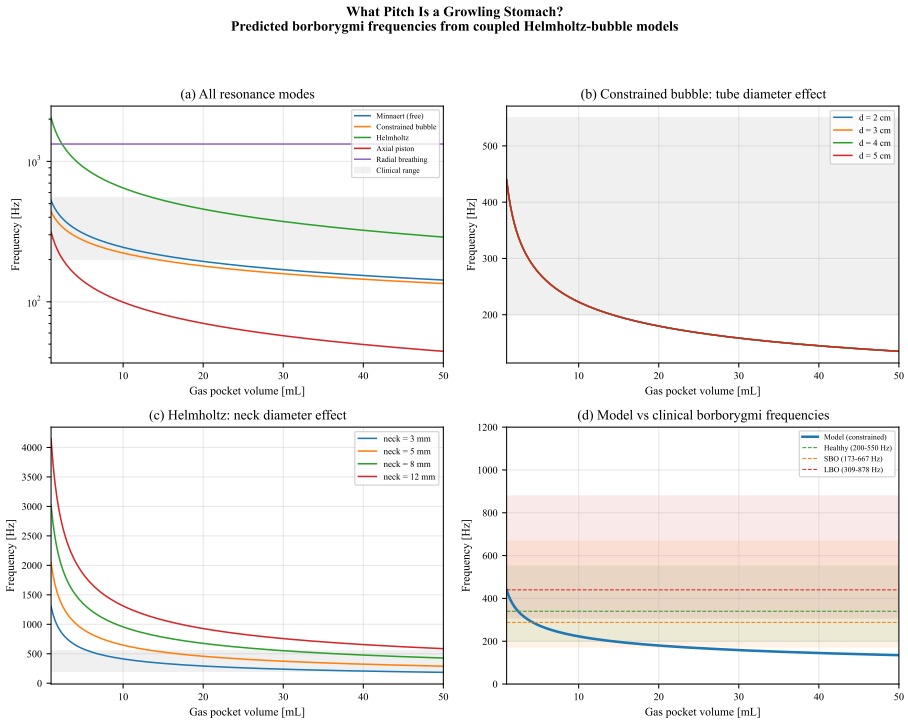

Figure 1: Predicted borborygmi frequencies as a function of gas pocket volume for the five resonance mechanisms. Shaded region: clinically observed healthy bowel sound range [2]. The constrained bubble model provides the best coverage of the clinical band.

Helmholtz mode consistently lies above both bubble modes, becoming relevant only when narrow constrictions are present.

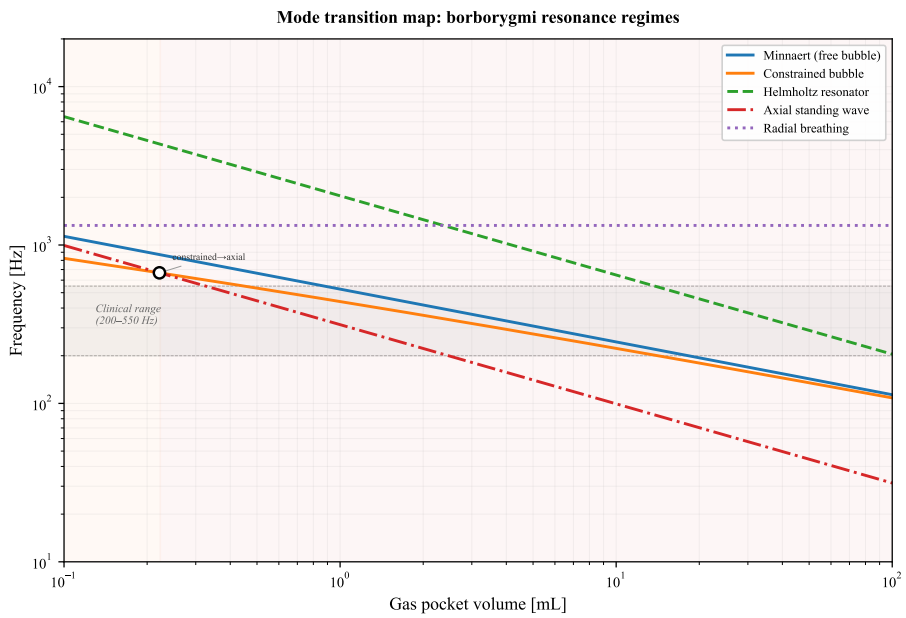

Figure 2: Mode transition map showing the five resonance mechanisms across gas pocket volumes from 0.1 mL to 100 mL. The clinical band (200 Hz to 550 Hz) is indicated by horizontal dashed lines. The constrained bubble mode dominates the clinically audible band; the axial piston mode has the lowest overall frequency above $V \approx 0.22$ mL but is sub-audible at physiological volumes.

3.4. Clinical comparison

Table 3 compares model predictions with reported clinical frequency bands. For each clinical condition, we identify the gas pocket volume that produces the reported peak frequency under the constrained bubble model.

The constrained bubble model predicts a range of 135 Hz to 440 Hz for 1 mL to 50 mL, which overlaps substantially with the healthy adult band of 200 Hz to 550 Hz reported by Ching and Tan [2]. The model's lower boundary (135 Hz at 50 mL) corresponds to the infrasonic rumbling that is felt but not heard, while the upper boundary (440 Hz at 1 mL) aligns with the peak frequency reported

Table 3: Comparison of constrained bubble model predictions with clinical borborygmi frequency data. Inferred volumes are obtained by inverting the constrained model at each reported peak frequency.

Condition	Reported range (Hz)	Peak (Hz)	Inferred V (mL)	Source
Healthy adult	200–550	340	∼3	[2]
Small bowel obstruction	173–667	288	∼5	[16]
Large bowel obstruction	309–878	440	∼1	[17]
Constrained model (1–50 mL)	135–440	—	1–50	This work

for large bowel obstruction [17].

The small bowel obstruction peak of 288 Hz [16] corresponds to a pocket volume of approximately 5 mL under the constrained model. This is physically plausible: small bowel obstruction involves proximal dilation with accumulated gas and fluid, producing pockets of several millilitres that oscillate at frequencies in the mid-range of the clinical band. The large bowel obstruction peak of 440 Hz [17] implies smaller gas pockets (∼1 mL), which may reflect fragmentation of gas into small volumes by colonic haustral contractions, or alternatively the stiffer colonic wall elevating frequencies through increased k_{wall}.

The model underpredicts the upper extent of the clinical range for healthy adults (550 Hz vs. the model's 440 Hz at 1 mL). This discrepancy is readily explained by sub-millilitre gas pockets—which are common in the small bowel [18]—or by Helmholtz resonance contributing to the high-frequency tail of the spectrum.

3.5. Parametric sensitivity

The constrained bubble frequency depends primarily on gas pocket volume V and secondarily on wall stiffness E and tube diameter d. We examined each parameter's influence through targeted sweeps.

Wall stiffness.. Increasing the wall modulus from 1 kPa (very soft mucosa) to 100 kPa (fibrotic bowel) raises k_{wall} proportionally. At 10 mL, however, the constrained frequency shifts only from approximately 221 Hz at $E = 1$ kPa to 241 Hz at $E = 100$ kPa—a modest 9% increase. The constrained frequency is thus remarkably insensitive to wall stiffness, dominated instead by gas compressibility. This insensitivity arises because k_{wall} is small relative to k_{gas} for the soft intestinal wall, even at a hundredfold increase in E. Pathological spectral shifts from fibrosis are therefore expected to be subtle, though potentially still detectable with sufficiently precise acoustic monitoring.

Tube diameter.. The constrained bubble frequency is independent of tube diameter, since the equivalent spherical radius R depends on volume alone and the constrained model (Eq. 4) contains no lumen-geometry terms. This is a limitation of the spherical approximation: in reality, the lumen boundary would modify the added-mass term for non-spherical pockets. However, tube diameter strongly influences the axial and radial modes, which scale directly with lumen radius R_l. For larger-diameter segments (colon, $d \approx 40$ mm to 60 mm), the radial breathing mode frequency decreases and may enter the audible range.

Neck geometry (Helmholtz).. The Helmholtz frequency is highly sensitive to neck diameter: doubling d_n from 3 mm to 6 mm approximately doubles the frequency at a given volume. This sensitivity means that the degree of peristaltic constriction modulates the pitch of click-type bowel sounds, consistent with the clinical observation that high-pitched tinkling sounds are associated with tight constrictions in obstruction.

4. Discussion

4.1. Why the constrained bubble model dominates

Among the five resonance mechanisms considered, the tissue-constrained bubble provides the best match to clinical observations for two reasons. First, typical intestinal gas pockets in the range of 1 mL to 50 mL are quasi-spherical

(or at least compact), so the spherical bubble assumption is reasonable. Second, the constrained model correctly captures the competing effects of wall inertia and wall stiffness. For the soft, dense intestinal wall ($E = 10\,\mathrm{kPa}$, $\rho_w = 1040\,\mathrm{kg/m^3}$), wall inertia slightly reduces the frequency below the free Minnaert value, while wall stiffness makes a comparatively small correction. The net effect places the constrained frequency within the clinical band without requiring any parameter tuning.

The Minnaert model alone overpredicts because it neglects wall inertia. The Helmholtz model is relevant only in the presence of a constriction neck, which is not the default geometry. The axial and radial modes operate in different frequency regimes (too low and too high, respectively) and require elongated or tube-filling gas column geometries.

4.2. Interpreting the mode transition map

The mode transition map (Fig. 2) reveals a nuanced landscape. By the lowest-frequency criterion, the axial piston mode dominates above $V \approx 0.22\,\mathrm{mL}$ because its frequency drops more steeply with volume ($f \propto V^{-1}$ vs. $V^{-1/3}$ for the constrained bubble). However, the axial frequency is sub-audible at physiological volumes ($f_{\mathrm{ax}} < 100\,\mathrm{Hz}$ for $V > 1\,\mathrm{mL}$), placing it below the clinically observable range. Within the audible band (200 Hz to 550 Hz), the constrained bubble mode is the dominant contributor. This distinction is clinically important: a single-mode model based on the constrained bubble is sufficient for interpreting auscultatory findings, even though the formal lowest-frequency mode is sub-audible.

The Helmholtz mode, while not dominant in the lowest-frequency sense, deserves attention as a secondary mechanism. When peristaltic contractions are present, the Helmholtz frequency may overlay the bubble frequency, producing the characteristic broadband, multi-tonal quality of real borborygmi. The mode transition map provides a quantitative framework for predicting which combinations of modes are active for a given pocket volume and constriction geometry.

4.3. Clinical implications

The model suggests a straightforward interpretation of clinical spectral shifts. Small bowel obstruction, which involves proximal dilation and accumulation of gas and fluid, produces moderate-sized gas pockets (3 mL to 10 mL) that resonate in the mid-frequency range (220 Hz to 340 Hz). The reported peak frequency of 288 Hz [16] corresponds to a pocket volume of approximately 5 mL under the constrained model—physically consistent with dilated small bowel.

Large bowel obstruction, by contrast, may involve either smaller gas pockets (due to colonic haustral segmentation) or a stiffer colonic wall. Either mechanism would elevate the frequency, consistent with the higher peak of 440 Hz reported by Du Plessis et al. [17]. The model's sensitivity to wall stiffness (Sec. 3.5) suggests that conditions affecting wall compliance—Crohn's disease, fibrosis, inflammation—could produce detectable spectral signatures.

If validated experimentally, the constrained bubble model could serve as a frequency inversion tool: given a measured peak frequency, one could infer the gas pocket volume, providing non-invasive information about intestinal gas distribution that is currently obtainable only through imaging.

4.4. Limitations

Several simplifications warrant discussion.

Geometry. Real intestinal gas pockets are irregular, not spherical or cylindrical. The spherical approximation is most accurate for compact pockets where the aspect ratio is near unity; elongated pockets would require a more detailed geometrical treatment. Furthermore, the equivalent-sphere diameter exceeds the lumen diameter for volumes above approximately 14 mL (a sphere of 14 mL has a diameter of 30 mm, matching the lumen). Larger pockets necessarily adopt elongated, non-spherical geometries to fit within the lumen. We therefore consider the spherical constrained-bubble model to be most reliable for volumes up to approximately 14 mL and treat predictions at larger volumes as an extrapolation that likely overestimates the effective bubble radius. The constrained

model is also strictly independent of tube diameter—a consequence of the spherical assumption that neglects lumen-boundary effects on the added-mass term.

Static equilibrium. We model steady-state resonance frequencies and do not account for the time-varying geometry imposed by peristaltic waves. Peristalsis modulates both the pocket volume and any constriction neck dimensions on timescales of seconds, producing the characteristic intermittent quality of borborygmi.

Single-pocket model. Real intestinal gas exists as multiple pockets distributed along the tract [19, 18]. Acoustic interactions between neighbouring pockets—analogous to coupled oscillator systems—could produce frequency splitting and beating phenomena not captured by the present single-pocket analysis.

Gas composition. We assume air ($\gamma = 1.4$) for the gas pocket. Intestinal gas is a mixture of nitrogen, carbon dioxide, hydrogen, methane, and hydrogen sulphide [20, 18], with a slightly different effective γ. For the dominant Minnaert and constrained modes, the frequency scales as $\sqrt{\gamma P_0}$, so modest changes in γ (from 1.4 to approximately 1.3 for a CO_2-rich mixture) would reduce frequencies by approximately 4%—smaller than the uncertainty in wall properties.

Damping. We use a single loss tangent $\eta = 0.25$ ($Q = 4.0$) to represent all dissipation mechanisms. In reality, viscous losses in the fluid, radiation damping, and viscoelastic losses in the wall contribute differently at different frequencies. The resulting quality factor of 4 is consistent with the short duration ($\sim$0.5 s to 0.8 s) of individual bowel sound bursts [2], though this comparison is approximate.

Radiation and transmission. We do not model how efficiently sound generated at the gas pocket propagates through tissue to the abdominal surface, where it is detected by stethoscope or microphone. This transmission path introduces frequency-dependent attenuation that would alter the measured spectrum relative to the source spectrum predicted here.

5. Conclusions

We have presented a unified analytical framework for predicting the resonant frequencies of intestinal gas pockets, comprising five distinct acoustic mechanisms: free Minnaert bubble oscillation, tissue-constrained bubble resonance, Helmholtz resonance, axial piston modes, and radial breathing modes. The principal findings are:

1. The tissue-constrained bubble model, which augments the classical Minnaert formula with intestinal wall elasticity and inertia, predicts frequencies of 135 Hz to 440 Hz for gas pocket volumes of 1 mL to 50 mL. This range overlaps the clinically observed healthy bowel sound band of 200 Hz to 550 Hz, specifically capturing the 200 Hz to 440 Hz sub-band without curve-fitting or fitted parameters. Frequencies above 440 Hz require sub-millilitre pockets or Helmholtz contributions.
2. A mode-transition map across the volume–frequency space shows that the axial piston mode has the lowest overall frequency above $\sim$0.22 mL, but it is sub-audible at physiological volumes ($<$ 100 Hz for $V > 1$ mL). The constrained bubble mode therefore dominates the clinically audible band across the entire physiologically relevant volume range. The Helmholtz mode provides a secondary high-frequency contribution relevant to obstruction-related tinkling sounds.
3. The model provides a physics-based framework for interpreting clinical spectral shifts: the reported peak frequencies for small bowel obstruction (288 Hz) and large bowel obstruction (440 Hz) correspond to pocket volumes of approximately 5 mL and 1 mL, respectively, under the constrained model.
4. Wall stiffness is a secondary parameter: a hundredfold increase (from 1 kPa to 100 kPa, representing fibrotic bowel) raises the predicted frequency by only $\sim$9%, indicating that the constrained frequency is dominated by gas compressibility rather than wall elasticity. Pathological spectral shifts from fibrosis, while potentially detectable, are expected to be subtle.

5. The spherical bubble assumption limits the model's high-confidence range to volumes below approximately 14 mL, above which the equivalent-sphere diameter exceeds the lumen diameter. Predictions for larger volumes should be treated as extrapolations.

Experimental validation through simultaneous bowel sound recording and imaging (to estimate pocket volumes) is a natural next step. If confirmed, the constrained bubble model could serve as a non-invasive frequency inversion tool for estimating intestinal gas pocket volumes from measured acoustic spectra.

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